Learner guide to marks and checked returns

Paper 000, version 0.1.0. AuthorityEffect: None.

The first game asks you to keep a question, a proposed answer, and its check distinct. This guide gives you a route through the paper and workbook, then prepares you for the Landau examination. Worked examination answers are in the separate instructor key. The Apartment 50 preface is optional fiction.

A route through the paper

Read and run	Learn to do	Quiz questions
Sections 1-5; workbook lessons 1 and 4	Count admitted marks; distinguish S(n), MSS-N, and execution; declare a board	1-5
Sections 6-8; workbook lessons 2-4	Supply an observation witness; state a notation's reading; distinguish types from values	6-10
Sections 7-10; workbook lessons 5-7	Rename consistently; bind the exact question; replay; stop honestly	11-15
Example 8 and Section 9; workbook lessons 6-8	Preserve a way back; distinguish repair from extension; audit an actual return	16-20
The additional definitions in Questions 21-24	Enumerate minima; certify a cycle optimum; declare cells; audit a consumer	Optional capstones

For each workbook cell, write down your prediction first. Run it, change one input, and explain any discrepancy. Keep the old input and result. A changed question should remain visibly different from a repaired answer.

The vocabulary you need

Term	Meaning in this introduction	A common mistake
Presentation	A supplied display or representation	Counting every printed character as an admitted mark
Carrier	The admitted marks on the declared board	Smuggling a new endpoint into an answer
Observation	A supplied value for each mark under a declared test	Treating different names as an observation witness
Goal	The finite property requested on this instance	Reversing the goal to make a failed answer pass
Candidate	A proposed answer, bound to a particular problem	Treating persuasive prose as a checked answer
Receipt	The scoped result of a declared checker	Extending CHECKED to claims outside its predicate
Return packet	Retained problem, candidate, and receipt	Trusting a detached status word
Renaming	A bijection applied consistently to structure	Collapsing different marks or changing only the answer
Replay	Recomputing the retained check	Assuming replay authenticates the producer
Extension	A declared change with new preservation obligations	Claiming an atlas successor from a local check

S(n) is a Genesis derivation stage; MSS-N is an external atlas specification; a session event is an execution record. They have different roles. The paper's stage table is orientation, not a new derivation or a license to advance Genesis.

What the checker does

For the basic relation exercise, think of the referee as following this finite procedure: check the field set and schema; validate the carrier and label; validate the goal and candidate shapes; check membership; reject self-pairs; compare labels; compare directed endpoints with the goal. A failed check returns its first reason. Later checks may not have run.

For the richer lesson packet, the declared task, input type, and answer type select the teaching predicate. English prompt text is retained data. The candidate

includes the exact problem identity. Editing the text can change that identity even when the finite predicate remains the same.

The current v0.1.0 profile checker has a small precedence discrepancy: on a table with no separator it reports BLOCKED before checking whether the answer value is an integer. Question 19 asks you to find and repair this. Observe the shipped behavior first. The proposed code patch makes malformed index values REFUSED before the missing-separator check.

Outcome	What it says	A useful next action
CHECKED	This value passed the declared finite predicate	Consume the checked value within scope; preserve the problem binding
REFUSED	This envelope or candidate failed a declared check	Inspect the reason; repair the answer or correct an invalid declaration
BLOCKED	The exercise lacks a required declared separator	Supply new input under a new problem; retain the old record
UNKNOWN	The bounded session obtained no checked answer	Retain attempts and describe the stopping condition

UNKNOWN covers both exhausted fuel and a proposer that ends without a passing answer. The current controller records attempts but has no separate stop-reason field. Do not claim the budget was exhausted if fewer attempts occurred.

A short worked prediction

On carrier p,q with goal p to q and label `separates`, the candidate `["q", "separates", "p"]` is a well-shaped irreflexive triple. It still misses the directed goal. The defect is not its syntax or a self-loop; it is the specific requested direction. This distinction makes the refusal useful.

Similarly, a two-column observation row `["red", "red"]` supplies no separator. The claim concerns that supplied row. It does not exclude every observation you could later declare. A different spelling for q changes neither value.

Read notation in its declared register

`<>` is one opaque presentation in the ground exercise and admits no mark. The paper separately supplies an atlas-style reading in which `<ab>` depicts a directed

pairing. Record that convention before using it. An obstruction display is not itself an independently verified obstruction certificate.

Bracket displays such as `<edge<p><q>>` and graph arrows such as p to q are different representations. The restricted bracket alphabet does not prohibit the JSON used in the paper. Balanced brackets alone do not define an AST, the semantics of `edge`, or an executable checker. This board's irreflexive relation rule also does not forbid the reflexive equality p=p.

Keep identity, replay, and trust separate

The teaching identity uses sorted dictionary keys and compact JSON. Key order does not matter under that profile; list order does. It is not an RFC 8785 canonicalization claim. The digest is not a signature.

Replay can catch a receipt changed without the corresponding input and check. It can also accept a whole self-consistent return for an unintended question. A consumer therefore needs an expected problem identity retained independently of the returned packet. It still trusts the selected checker and needs separate evidence if producer authentication matters.

Hints for the examination

Try without these first. A hint points to an operation, not a verdict.

Questions	Hint
1-3	Name what each count or index refers to; distinguish external supply from admission
4-5	Follow the actual validation order rather than listing every possible defect
6-7	Compare values within each observation row; inspect exact Python types
8-10	Separate typography, answer metadata, value shape, and the selected predicate
11-12	Structural correspondence and exact-content identity answer different questions
13-14	Keep the caller's expected question outside the untrusted return
15	Count calls to `next`, not merely values already written in a list

Questions	Hint
16-17	Check old-image terms first; track which edits change the prompt
18-19	One counterexample suffices to refute a universal claim; actual code outranks your prediction
20	Choose a predicate another person can check completely
21	Write each mark's selected-row signature; retain ties rather than the first success
22	Sum differences around the cycle; then place residual on a least-weight edge
23	List omitted vertices and cancel oriented edge terms explicitly
24	A return can be internally consistent and answer the wrong question

Additional mathematics for the capstones

In Question 21, two marks are distinguished exactly when their selected observation signatures differ. There are 2^k binary signatures of length k. This is a finite capacity bound under the declared binary model, not a theorem about native Genesis growth.

For Question 22, each oriented edge uv has difference f(v)-f(u). Residual is the report minus that difference. Weighted absolute residuals give the cost. If a circulation lambda has zero divergence, then lambda dot df=0: each vertex potential has equal incoming and outgoing coefficients. With |lambda_e| <= w_e, termwise domination gives

[c= (c-df) _e w_e|c_e-(df)_e|.]

A candidate cost matching this lower bound proves an optimum for that instance. No numerical optimizer or unpriced approximation is needed for the cycle.

Question 23 explicitly supplies the ordered simplex and horn conventions. The standard definition of a horn is given in Kerodon, Subsection 1.2.4. A drawing of a graph does not admit higher cells; you must declare which faces exist and which target simplex you mean. Do not infer physics from a classroom chain calculation.

Run the local examples

From the extracted paper directory, with Python 3.12 or later and the companion dependency configuration corrected as described in the review:

```
python -m pip install ".[dev]"
python code/basic_packets.py
python code/lesson_packets.py
python -m pytest code/tests -q
python verify_deposit.py
```

On Windows, `py -3.12` may replace `python`. The two teaching scripts use only the standard library. Tests need pytest. Notebook execution needs nbformat, nbclient, and ipykernel; a notebook interface must also be available if you want to use a browser UI. `verify_deposit.py` checks the manifest unless explicit test flags are supplied. It does not prove all manuscript claims.

The supplied release archive check requires the original ZIP to be present inside the extracted paper folder. It is an optional maintainer check, not part of running the teaching examples.

After the core examination, continue to Paper 001 for stronger exact graph certificates and Paper 005 for session management. These are routes to further work, not material assumed by the core questions.